Week 4 Decision Gate: *King Octopus and the servants*

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Source and Problem Statement

The following puzzle is from Puzzle Prime:

King Octopus and His Servants

King Octopus has servants with 6, 7, or 8 legs. The servants with 7 legs always lie, and the servants with 6 or 8 legs always tell the truth. One day four of the king's servants had the following conversation:

> "We together have 28 legs," said the first octopus. "We together have 27 legs," said the second octopus. "We together have 26 legs," said the third octopus. "We together have 25 legs," said the fourth octopus.

Question: Which of the four servants told the truth? *Source:* Puzzle Prime, "King Octopus and His Servants."

Rules / Assumptions

What the key terms mean here: *valid*, *follows*, "for every integer", a servant has exactly one of $6, 7, 8$ legs, ...

Work

Claim. *State the claim precisely.*

Proof. Steps, each justified by a definition or a prior step. Or, if the claim is false: □

Counterexample. *An object in the claim's domain that satisfies the hypothesis and violates the conclusion. Show the values.*

Check Your Answer

Either verify *every* proof step against a definition, or verify the counterexample really satisfies the hypothesis and violates the conclusion. For an argument-validity problem, exhibit premises that are all true while the conclusion is false (or argue that is impossible).

One-Sentence Summary

Exactly what you established — and, where it matters, distinguish “the conclusion happens to be true” from “the argument is valid.”

Strongest disagreement

The best objection a sharp, skeptical reader would raise, stated honestly, and whether it survives.